

From Recreational to Elite: Performance Stratification in Marathon Results

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Abstract

Marathon results are often reduced to a single average or a familiar target such as sub-3 or sub-4 hours. This study uses 429,266 finisher records from 641 U.S. marathons in 2023 to visualize the full performance structure of the sport. We combine runner-level finish time, age, and gender with race-level size and date. The median finish time was 4.46 hours; 4.5% of runners finished under 3 hours and 33.9% finished under 4 hours. Performance composition varied strongly by age and race size, while race size alone did not explain competitiveness. October and November contained 44.2% of listed finishers. The visual analysis shows why marathon performance should be interpreted across both demographic and race contexts rather than through one threshold or ranking.

I. Introduction

A marathon result is more than a single finishing time. A sub-4-hour result may represent a strong recreational performance, but its meaning changes with the runner's age, gender, and race context. A race with tens of thousands of finishers can also have a different field from a small local event. Simple averages hide these differences.

This project studies *performance stratification*: how finishers are distributed from near-elite to recreational levels. The main contribution is a set of coordinated static visualizations that moves between two scales. The runner-level view explains the overall distribution and demographic differences. The race-level view explains how field size, calendar timing, and the mix of faster runners vary among events.

The study answers four questions:

1. How are marathon finishers distributed across interpretable performance bands?
2. How do finish-time distributions and performance-band shares vary by age and gender?
3. How does race size relate to the performance composition of an event?
4. Which months and races contain unusually large or competitive fields?

The project is exploratory rather than predictive. It does not rank individual runners or claim that race context causes performance. Instead, it uses visual comparison to reveal patterns that are difficult to see in raw result tables.

II. Previous Work

Research on endurance running shows why demographic context matters. Leyk et al. found that age-related decline in marathon and half-marathon performance is gradual through much of adulthood and becomes stronger at older ages [1]. Later studies of the New York and Berlin marathons also found changing participation and nonlinear age-performance patterns across large fields [2], [3], [4]. Work on elite marathoners further shows that age and sex interact with the performance level being studied [5]. These findings support comparing complete distributions across age groups instead of treating age as a minor control variable.

Finish time also reflects factors that are absent from our dataset. Smyth's analysis of 1.7 million recreational runners showed that pacing decisions differ within races and can materially affect the final result [6]. A longitudinal study of runners who maintained sub-3 performance across five decades also showed that highly trained individuals do not follow the same simple decline implied by a population average [7]. These studies motivate cautious interpretation of fixed performance bands: the bands are useful visual summaries, but not complete measures of ability.

The project also follows established principles of quantitative display. Few argues that a chart should make its intended comparison direct and should avoid decoration that competes with the data [8]. We apply this idea by using position and length for the main comparisons, consistent finish-time thresholds across figures, restrained color, and direct labels where space permits. Histograms show distribution shape; point ranges show medians and interquartile ranges; heatmaps show repeated part-to-whole comparisons; and scatterplots preserve race-level variation that an average would hide.

Most public marathon summaries focus on winners, qualifying times, or one race at a time. The gap addressed here is a multi-scale view of the entire 2023 field. The same performance bands are followed from the population level to demographic groups and then to race contexts. This makes it possible to see both the broad structure and the exceptions.

III. Dataset

A. Data Source and Structure: The primary source is the 2023 Marathon Results dataset [9]. **Results.csv** contains

Table I: Dataset and quality summary

Measure	Count
Runner-level result records	429,266
Race metadata records	641
Finishers listed in race metadata	429,982
Implausible or unavailable ages	8,779
Gender X or U records	392

429,266 runner-level records with **Race**, **Year**, **Gender**, **Age**, **Finish**, and **Age Bracket**. **Races.csv** contains metadata for 641 races, including **Date** and **Finishers**. We joined the files by race name and year. All finish-time calculations use the runner-level **Finish** field, recorded in seconds.

The analysis is cross-sectional. It compares races and finishers within 2023 and cannot establish long-term trends. Runner names are identifiers in the source data but are not used in the analysis or visualizations.

B. Data Quality: Table I shows the main data checks. Race metadata lists 716 more finishers than appear in the result file, a difference of 0.17%. We therefore use result rows for runner-level percentages and use the metadata total only in the calendar view. Ages below 10, including the source value of -1, are treated as unavailable in age-specific analysis. The small X and U gender categories are retained in overall totals but are not shown as separate demographic distributions because their combined count is only 392. Finish times are not removed as outliers; the focused histogram changes only the visible x-axis range, not the band counts.

IV. Methodology

A. Processing and Derived Variables: The analysis was implemented in R 4.5.2 with **readr**, **dplyr**, **tidyr**, **ggplot2**, **scales**, and **cowplot**. The reproducible script loads both CSV files, converts finish time from seconds to hours, parses race dates, joins race metadata, computes summaries, and exports vector PDF figures.

We created five mutually exclusive performance bands:

- under 2:45, representing elite or near-elite results;
- 2:45-2:59, completing the sub-3 group;
- 3:00-3:59;
- 4:00-4:59; and
- 5:00 or slower.

These thresholds are familiar and easy to read, but they are descriptive rather than official ability categories. They do not adjust for age, gender, course difficulty, or weather. The optional Boston Cutoff Time Tracker dataset could later support an age- and gender-adjusted comparison [10], but it is not used in this version.

Race size is divided into four groups: small (fewer than 100 finishers), medium (100-499), large (500-1,999), and major (2,000 or more). For each race, we calculate the median finish time and the shares finishing under 3 and under 4

hours. Race-level scatterplots require at least 25 records, while the competitive-race ranking requires at least 500. These thresholds reduce unstable percentages from very small fields.

B. Visual Design: The figures use the same performance-band order and colors throughout the paper. Sequential values use a single light-to-dark scale. Position and bar length carry the main quantitative comparisons, so the meaning does not depend on color alone. Counts and percentages are directly labeled when this improves reading. Multiple panels are aligned on a common visual structure to support comparison without an interactive dashboard.

The plots are descriptive. The LOESS line in the race-size scatterplot summarizes the visible pattern but is not a fitted causal model. Likewise, race rankings describe the participants recorded in this dataset; they do not measure course difficulty or event quality.

V. Results

A. Overall Performance Distribution: Figure 1 shows a right-skewed finish-time distribution. The median is 4.46 hours, and the largest single band is 4:00-4:59 with 144,271 finishers (33.6%). A similar 139,308 finishers (32.5%) took 5 hours or more. Faster results are much less common: 19,357 runners (4.5%) finished under 3 hours, including 5,449 (1.3%) under 2:45. Overall, 33.9% finished under 4 hours and 67.5% finished under 5 hours. The histogram displays 2 to 8 hours for readability, while the bar chart includes all records.

B. Age and Gender Stratification: Figure 2 compares medians, interquartile ranges, and performance-band shares for women and men with known age brackets. Finish times generally become slower in older groups, but the chart also shows the spread within each group. Among runners under 35, the median is about 4.58 hours for women and 4.16 hours for men. The difference narrows at older ages, while uncertainty grows because the oldest groups contain fewer runners.

The heatmap adds information that a median cannot show. For example, 34% of women under 35 and 23% of men under 35 are in the 5+ band. By ages 65-69, these shares rise to 67% and 52%, respectively. The 75-79 and 80+ cells should be interpreted cautiously because their sample sizes are much smaller. The result is therefore a distributional pattern, not a claim about any individual runner.

C. Race Size and Performance Composition: The dataset contains many more small events than major marathons. Figure 3 shows 314 small races, 210 medium races, 88 large races, and 29 major races. However, the performance mix does not change in a simple linear way. The 5+ band contains 53.0% of runners in small races, falls to 31.3% in medium races and 24.8% in large races, then rises to 34.1% in major races. Large races have the highest combined share between 3 and 5 hours, while major races contain

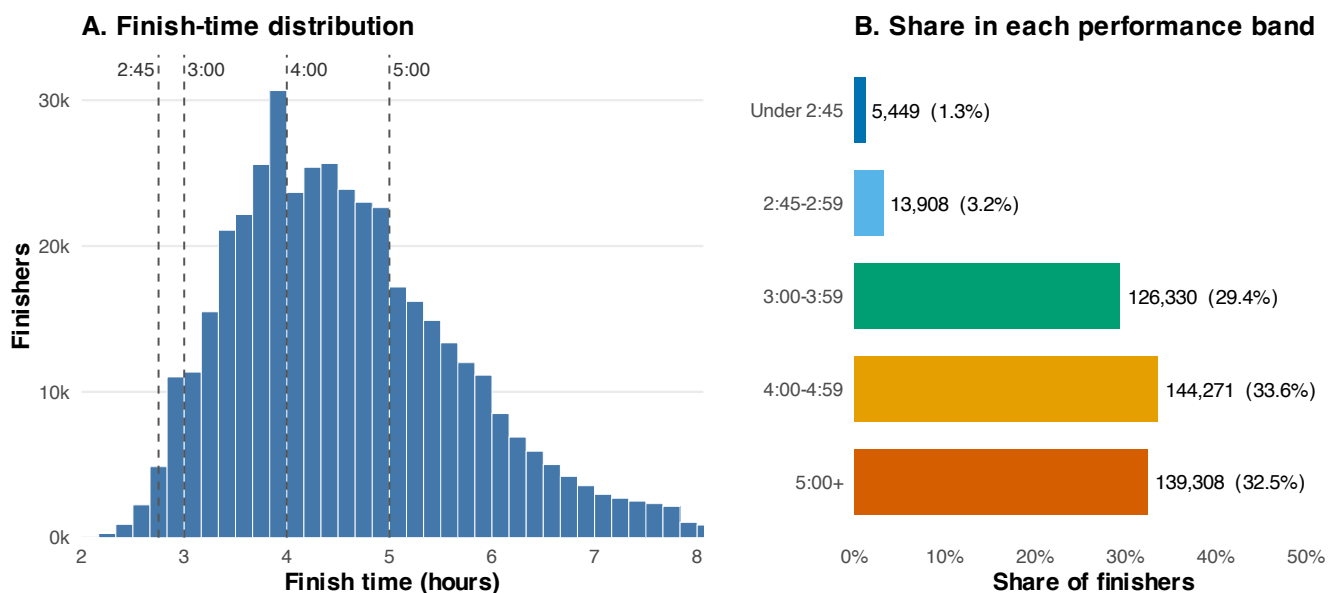
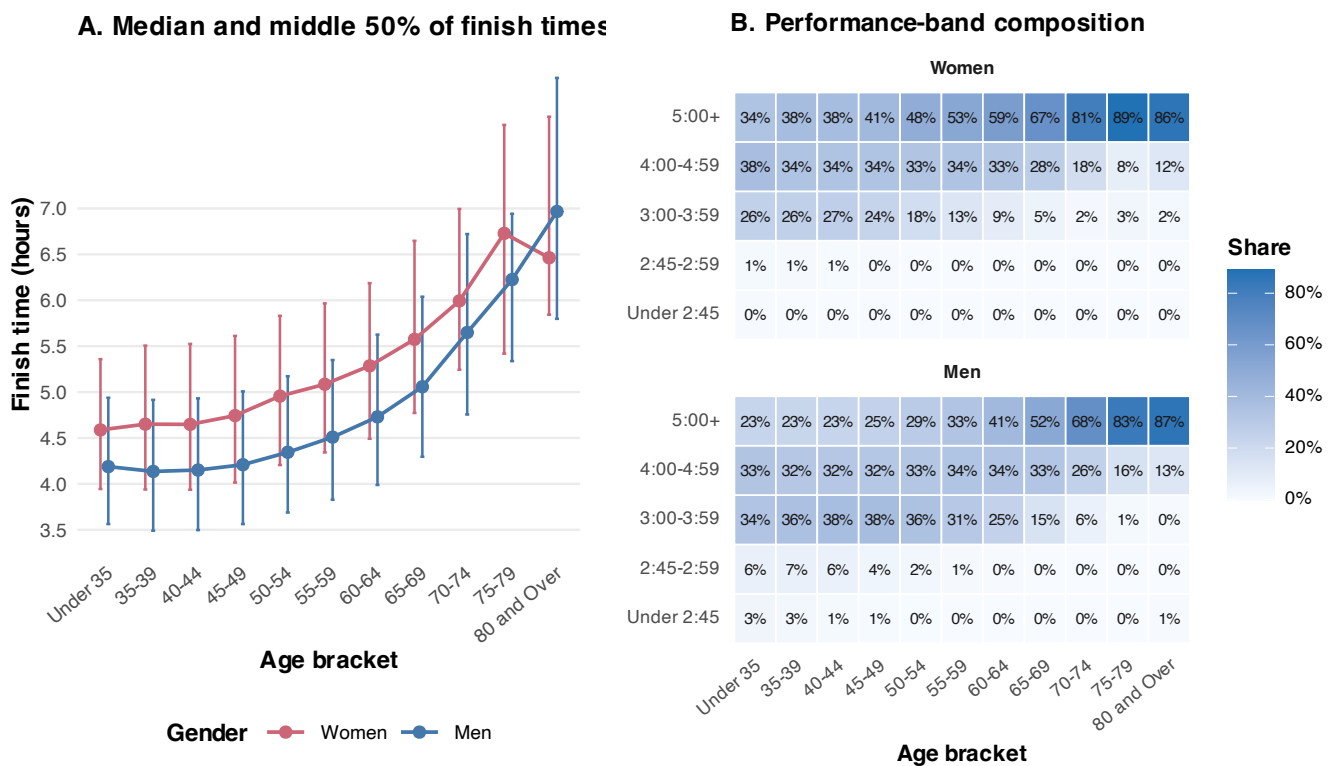


Figure 1: Overall performance distribution. Dashed lines mark the fixed band thresholds; band counts include all 429,266 result records.



local event type and a limited number of participants.

Figure 4 preserves the variation among 576 races with at least 25 records. The trend rises from very small to medium-sized fields, then levels off among larger events. Variation remains wide at almost every size. Boston is an important exception, with 72.5% of its field under 4 hours, consistent with its qualifying entry system. Chicago is near 43%, while New York City is near 28%. These examples demonstrate that event selection and participant mix matter more than size alone.

D. Race Calendar: Race count and participant volume are not distributed evenly through the year. Figure 5 shows that October has the most races (97) and the most listed finishers (107,590). November has only 53 races but 82,342 finishers because it includes several major events. In contrast, September has 82 races but only 14,693 finishers. October and November together account for 44.2% of the finishers listed in race metadata. The calendar is therefore not only seasonal; a few large autumn races strongly shape the dataset.

E. Race-Level Competitiveness: Figure 6 ranks races with at least 500 records by sub-3 share. California International Marathon (15.9%) and Boston Marathon (15.7%) lead the ranking, followed by Light at the End of the Tunnel Marathon (11.1%) and Sugarloaf Marathon (10.7%). The top group includes both very large races and smaller events known for attracting time-focused runners. This supports the earlier result: field size and competitive concentration are related in some cases, but they are not interchangeable measures.

The ranking is deliberately based on a share rather than a raw count. A count would mainly reproduce race size. The minimum-field rule also prevents a race with only a few finishers from reaching the top because of one fast result.

F. Robustness and Interpretation Checks: We performed three checks to separate stable findings from choices made for visualization. First, the overall sub-4 share is 33.9% when each runner has equal weight. In contrast, the median sub-4 share among the 576 races with at least 25 records is 26.1% when each race has equal weight. This difference is expected because large races contribute many more runners to the population result. It also shows why the paper reports both participant-level and race-level views rather than treating them as the same quantity.

Second, the Spearman correlation between race size and sub-4 share is 0.53 among races with at least 25 records. This is a moderate positive association, consistent with the upward part of the trend in Fig. 4. However, the wide vertical spread and the lower values for some of the largest races show that size is not sufficient for prediction. The correlation supports a broad tendency, while the scatterplot preserves the exceptions.

Third, we repeated the sub-3 ranking with minimum field sizes of 250, 500, and 1,000. These cutoffs leave 195, 117,

and 67 eligible races, respectively. Seven of the top 10 races under the 500-runner rule remain in the top 10 under the 1,000-runner rule. California International Marathon and Boston Marathon remain the top two under all three cutoffs. Lower ranking positions change because smaller time-focused races enter or leave the eligible set. We therefore treat the first two positions as stable in this dataset and the remaining order as threshold-sensitive.

The demographic and calendar conclusions are also unlikely to be driven by small data-quality groups. Implausible or unavailable ages make up 2.0% of result rows and are excluded only from age-specific views. X and U together make up 0.09% and remain in overall totals. The 0.17% difference between result rows and listed race finishers is too small to explain the large October and November concentration, although it prevents exact reconciliation of the two files. These checks support the main visual findings while keeping their limits explicit.

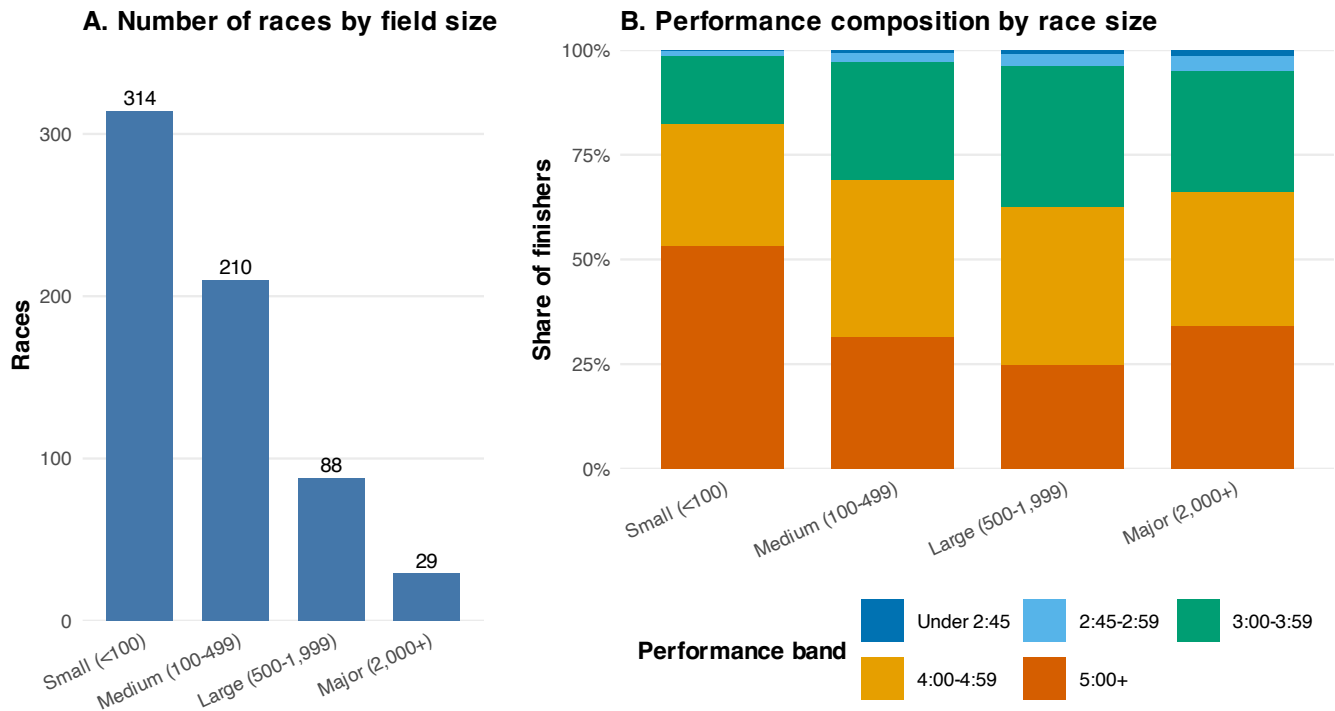


Figure 3: Race-size context. Panel A counts events in each field-size group. Panel B compares the within-group composition of finish-time bands.

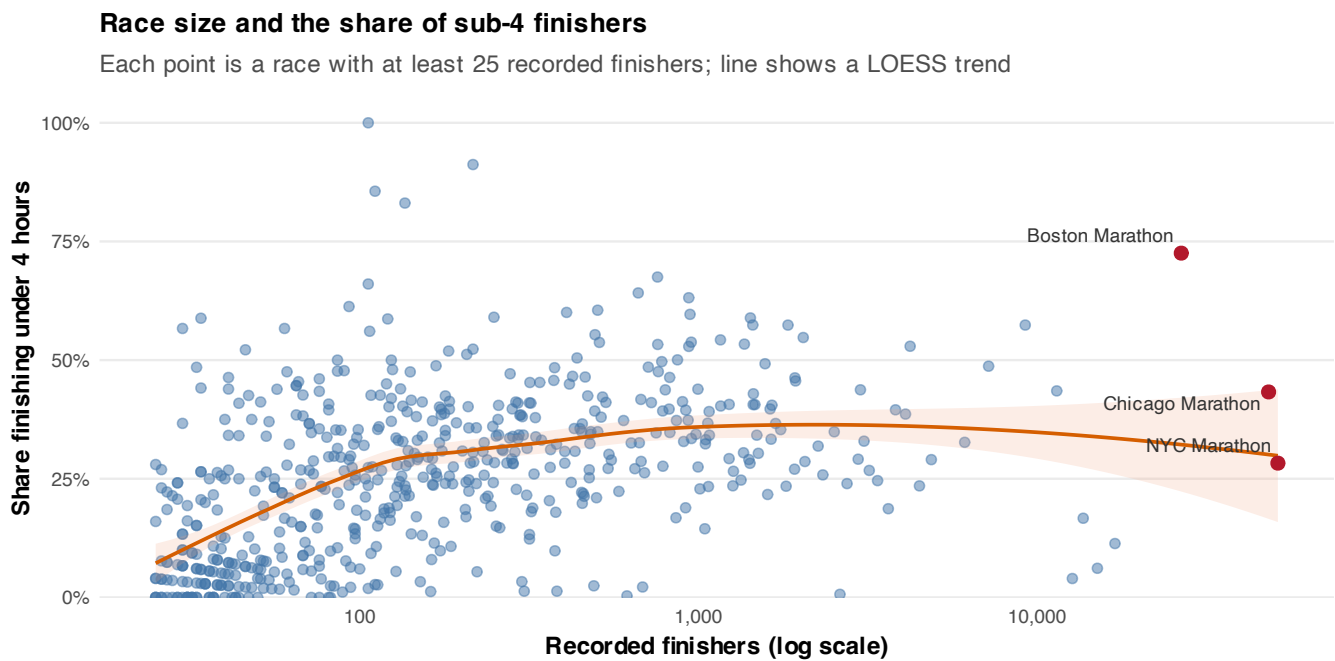


Figure 4: Recorded race size versus the share of sub-4 finishers for races with at least 25 results. The log x-axis keeps small and major events visible together.

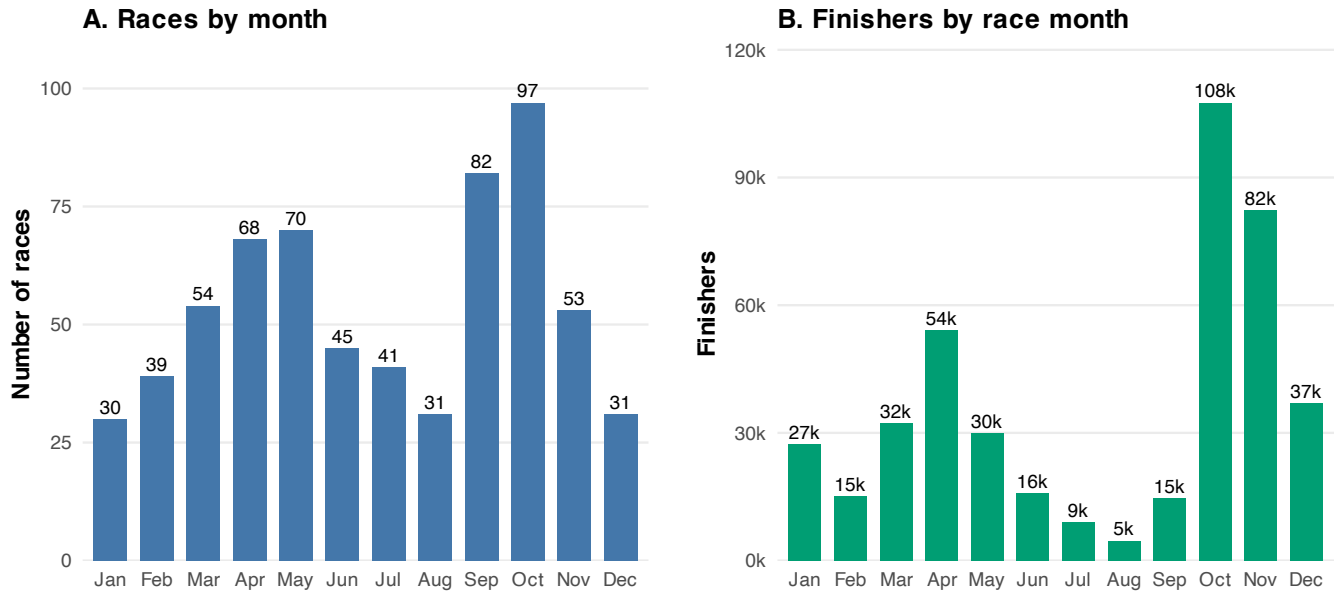


Figure 5: The 2023 race calendar. Panel A counts races, while Panel B sums the finishers listed in race metadata.

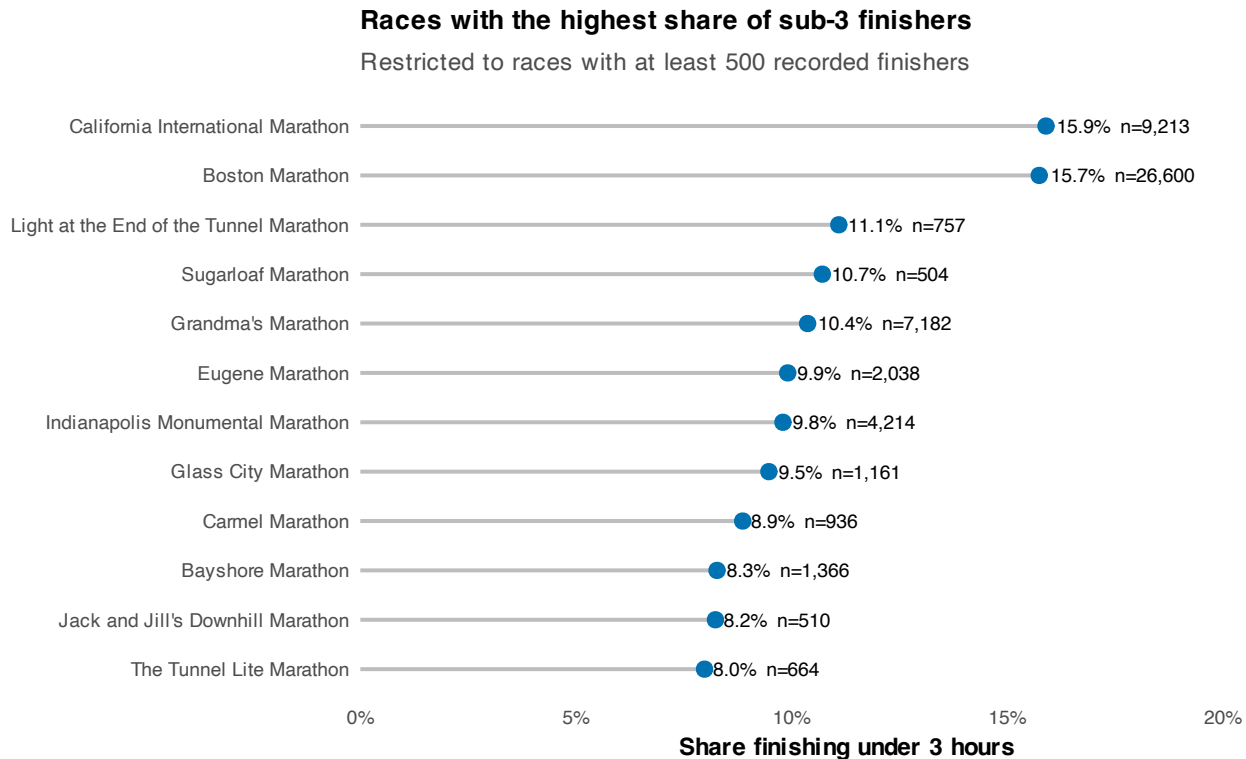


Figure 6: The 12 highest sub-3 shares among races with at least 500 recorded finishers. Labels show the percentage and denominator.

VI. Discussion

The figures answer the research questions at three connected levels. First, marathon performance is strongly concentrated in the recreational range: two-thirds of finishers take at least 4 hours, while sub-3 results form a small tail. Second, age and gender change both the center and composition of the distribution. Third, race context changes what the overall threshold means. Small, large, and major events have different mixtures, and races of similar size can have very different sub-4 shares.

The combination of views is more useful than one race ranking or one population average. The population chart gives scale, the demographic chart explains subgroup structure, and the race charts show where the overall pattern does not hold. Using the same bands across these views creates a visual link between individual and event-level analysis.

The most distinctive part of the project is the use of race metadata as analytical context rather than treating **Race** as only a label. Field size, date, and race-level performance shares reveal autumn concentration, broad-field major events, and highly competitive exceptions. This directly expands the analysis beyond performance level, age, and gender.

VII. Limitations

The dataset covers only U.S. marathons in 2023, so the findings cannot establish long-term trends. It does not include weather, elevation, course profile, split times, training history, or entry rules. These missing variables may explain some race-level differences. Boston's qualifying system, for example, helps explain its unusual position, but comparable entry-policy fields are unavailable for all races.

Fixed performance bands are easy to understand but do not adjust for age and gender. Race-level percentages are also descriptive and can remain unstable even after minimum-size filters. Finally, the 716-record difference between race metadata and runner results means calendar totals and runner-level totals are not exactly identical. Future work could add Boston qualifying standards or age grading, validate race metadata from event sources, and compare multiple years.

VIII. Conclusions

This project visualized 429,266 results from 641 U.S. marathons in 2023. The median finish time was 4.46 hours; 4.5% of finishers ran under 3 hours and 33.9% ran under 4 hours. Older age groups had progressively larger 5+ shares. Race context also mattered: small races had a 53.0% share of 5+ finishers, while large races had 24.8%, but major events again showed a broader participant mix. October and November dominated participant volume, and California International Marathon and Boston had the highest sub-3 shares among races with at least 500 records.

These findings show that no single threshold, average, or race size describes marathon performance well. A coordinated set of distribution, demographic, calendar, and race-level views provides a clearer account of how recreational and competitive performance coexist in the same dataset.

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